

Math 125 Final Review 2

Solutions on the following page.

1. Spring 2019 Q6
2. Fall 2018 Q7
3. Fall 2015 Q10

1. Q1

$$(a) \ V = (3\pi\sqrt{\frac{6}{5}} - \frac{5}{6}(\frac{6}{5})^{\frac{3}{2}}) = \sqrt{\frac{6}{5}}2\pi \approx 6.88$$

$$\text{Other forms: } V = \frac{2}{5}\pi\sqrt{30}$$

$$V = \frac{2}{3}\pi(\sqrt{\frac{6}{5}})^3 + \frac{6}{5}\pi(\frac{3}{\sqrt{\frac{6}{5}}} - \frac{3}{2}(\sqrt{\frac{6}{5}}))$$

2. Q2

$$(a) \ \frac{1}{e}$$

3. Q3

$$(a) \ V = 400\sqrt{\frac{400}{\pi}} = 8000/\sqrt{\pi} \approx 4513$$